Project Proposal: SEFC-Net (Self-Evolving Federated Cognitive Network)

A Next-Generation Federated Learning Architecture for Autonomous, Evolving, and Cognitively Intelligent Edge Systems

1. Project Overview

Title:

SEFC-Net: A Self-Evolving Federated Cognitive Network for Distributed Autonomous Intelligence

Principal Investigator / Student:

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Institution:

University of Rwanda, ACEIoT

Research Domain:

Artificial Intelligence, Federated Learning, Edge Computing, Reinforcement Learning, Cognitive Systems, Reconfigurable Intelligent Surfaces (RIS)

Duration:

12–18 months (extendable to PhD)

2. Abstract

Modern distributed systems — from smart cities and drones to civic sensors and health devices — suffer from limited autonomy and adaptability. Traditional federated learning allows decentralized training without data sharing, but it remains static, slow to adapt, and dependent on central orchestration.

SEFC-Net (Self-Evolving Federated Cognitive Network) introduces a **new class of intelligent architecture** that fuses **Federated Learning (FL)**, **Meta-Learning (ML)**, and **Reinforcement Learning (RL)** into a **self-evolving cognitive framework**. This network continuously learns, adapts, and optimizes itself across distributed IoT nodes — even in dynamic, bandwidth-limited, or resource-constrained environments.

The system mimics *biological evolution* — where each edge device (a "neuron" in a digital organism) learns locally, collaborates globally, and evolves intelligently.

3. Problem Statement

Despite rapid advances in IoT and AI, current distributed systems face critical limitations:

Challenge	Description	Limitation
Centralized Control	Traditional AI relies on a central model server	Scalability bottleneck, privacy risk
Static Learning	Models don't adapt after deployment	Poor performance in dynamic contexts
Low Autonomy	Edge devices are passive learners	High latency, low self-optimization
Data Privacy & Fragmentation	Civic and health data are siloed	Weak collective intelligence

Hence: We need a **federated, adaptive, and self-evolving intelligence** that enables devices to *learn, evolve, and cooperate* without central dependence — enabling smart cities, drones, and health systems to become *truly autonomous.*

4. Research Objectives

General Objective

To design and implement a **Self-Evolving Federated Cognitive Network (SEFC-Net)** that enables distributed IoT devices to learn autonomously, evolve their models, and adapt to new environments without centralized control.

Specific Objectives

1. Develop a **federated meta-learning framework** that allows each node to learn from limited data and adapt quickly to new tasks.
2. Integrate **reinforcement learning** for autonomous decision-making and reward-based model evolution.
3. Implement a **neuro-evolutionary optimizer** that mutates model architectures based on performance feedback.
4. Simulate IoT environments (sensors, drones, health devices) using Python-based edge clients.
5. Evaluate SEFC-Net's adaptability, scalability, and energy efficiency under dynamic environments.
6. Visualize real-time network evolution through an interactive **AI dashboard.**

5. Research Questions

1. How can we combine federated, meta-, and reinforcement learning into a unified, self-evolving system?
2. What mechanisms enable edge devices to autonomously optimize without centralized control?
3. Can a cognitive network adapt to changing data environments faster than static federated models?
4. How can privacy, scalability, and energy-efficiency be maintained in self-evolving architectures?

6. Methodology

6.1 System Architecture Overview

SEFC-Net Architecture Layers:

1. **Local Cognitive Clients (IoT Nodes):**
   Each node runs local models (CNN/MLP) and reinforcement agents that adapt based on environment feedback.
2. **Federated Meta-Learner:**
   Aggregates model updates, but unlike FedAvg, uses **reward-weighted dynamic aggregation** and **meta-optimization**.
3. **Self-Evolving Engine (Core Innovation):**
   A neuro-evolution module mutates and replaces underperforming models dynamically, guided by reinforcement rewards.
4. **Edge Simulation Layer:**
   Simulates drones, wearables, or health sensors generating live data streams (using MQTT or Node-RED).
5. **Dashboard Visualization:**
   Real-time visualization of model evolution, adaptation rates, and federated communication graph using **Dash/Plotly**.

6.2 Algorithmic Components

Component	Description	Tool / Framework

Federated Learning	Privacy-preserving distributed model training	TensorFlow Federated / PySyft
Reinforcement Learning	Adaptive model evolution & exploration	Stable-Baselines3 / RLlib
Meta-Learning	Model learns to learn (MAML-like updates)	TorchMeta / Learn2Learn
euro-Evolution	Model architecture mutation & self-growth	DEAP / NEAT-Python
Optimization & Auto-tuning	Auto-parameter optimization	Optuna
Visualization	Real-time evolution and metrics	Dash / Plotly / FastAPI

6.3 Data Sources

Use or simulate:

- Civic mobility datasets (e.g., CityPulse, OpenStreetMap mobility)
- Healthcare data (e.g., PhysioNet, wearable sensor data)
- Financial transaction logs (for SACCO-like simulations)
- Synthetic data for IoT edge behavior

6.4 Evaluation Metrics

Metric	**Meaning**
Accuracy	Local and global model performance
Adaptation Time	Time taken to learn new tasks
Energy Efficiency	Computation cost on edge devices
Privacy Score	No raw data transfer verified
Communication Overhead	Federated bandwidth usage
Evolution Rate	Model improvement per generation

7. Expected Results

- A **fully functional SEFC-Net prototype** simulating federated, meta-, and reinforcement learning on distributed IoT nodes.
- Demonstration of **self-evolving AI** that improves without central re-training.
- Visual dashboards showing model evolution and reward trends.
- Publication-ready findings demonstrating superior adaptability and privacy.
- Foundation for **PhD research** or **startup prototype (CivicSignal OS / Aetha Cloud).**

8. Novelty and Significance

Feature	**Existing Systems**	**SEFC-Net Advantage**
Learning Type	Federated or centralized	Federated + Meta + RL hybrid
Adaptation	Manual tuning	Self-tuning via RL rewards
Privacy	Basic differential privacy	Fully decentralized cognition
Scalability	Limited by central server	Fully distributed, node-level intelligence
Intelligence	Static	Self-evolving and cognitive
Real-world Impact	Academic or niche	Scalable civic, health, and mobility systems

9. Potential Applications

Sector	**Use Case**
Smart Cities	Adaptive traffic lights, crowd analytics
Healthcare	Decentralized patient monitoring and diagnostics
Finance / SACCOs	Behavioral learning for community finance
Drones / Mobility	Cooperative drone fleets and swarm optimization
Agriculture	Self-learning farm sensors and climate adaptivity

10. Tools & Technologies

Layer	**Technology**
Programming	Python 3.10, PyTorch, TensorFlow
Federated Learning	TensorFlow Federated, PySyft

RL / Meta-Learning	Stable-Baselines3, Ray RLlib
Evolutionary Algorithms	DEAP, NEAT-Python
Simulation	Node-RED, paho-MQTT, SimPy
Visualization	Dash, Plotly, TensorBoard
Version Control	Git + GitHub
Deployment	Docker + Edge simulation scripts

11. Expected Impact

- **Scientific:**
  Introduces the world's first *Self-Evolving Federated Cognitive Network*.
- **Social:**
  Empowers smart cities and health systems to evolve autonomously with privacy.
- **Economic:**
  Enables low-cost AI for developing regions through federated civic networks.
- **Academic:**
  Opens multiple publications in IEEE IoT, Nature Machine Intelligence, and ACM Transactions on Cyber-Physical Systems.

12. Future Work / PhD Expansion

PhD Topic Suggestion:

"RIS-Enhanced Self-Evolving Federated Cognitive Systems for Scalable Autonomous Intelligence"

Expands SEFC-Net by integrating:

- **RIS (Reconfigurable Intelligent Surfaces)** for intelligent wireless optimization
- **Quantum-assisted learning nodes**
- **Cross-domain federated cognition** across multiple infrastructures

13. Deliverables

Phase	**Deliverable**
1	Prototype of federated meta-learning system
2	RL-driven self-evolving optimization engine

Phase	Deliverable
3	Dashboard visualization + simulation demo
4	Research paper submission
5	Open-source release + funding proposal

14. Conclusion

SEFC-Net proposes a **new AI evolution paradigm** — one that bridges the gap between intelligence and autonomy. By combining federated learning, meta-learning, and reinforcement learning, it sets the foundation for living, adaptive, privacy-preserving systems that learn from communities rather than control them.

This project aligns with global research trends in **Edge AI**, **Cognitive Networking**, and **AI for Social Good**, making it both technically profound and societally transformative.

15. References

1. Kairouz, P. et al. (2021). *Advances and Open Problems in Federated Learning*. Foundations and Trends® in Machine Learning.
2. Finn, C. et al. (2017). *Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks (MAML)*. ICML.
3. Li, T. et al. (2020). *Federated Optimization in Heterogeneous Networks*. Proceedings of MLSys.
4. Stanley, K. O., & Miikkulainen, R. (2002). *Evolving Neural Networks through Augmenting Topologies*. Evolutionary Computation.
5. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press.